

TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models

Daniel Kushnir

Hao Wang

Abstract

Time-series foundation models (TSFMs) are deployed zero-shot, yet the tools that made language models inspectable do not transfer: tokens span different lengths of time across architectures, no ground truth defines a forecasting “concept”, and causal effects of internal features are small enough that uncontrolled measurements produce confident false positives. We present TSFM-LENS, an open-source system that turns a list of checkpoints into one self-contained interpretability report with one command. It is controlled: a leakage-audited synthetic benchmark with known generative components validates every probe. It is causal: concepts are defined by what removing them does to the forecast, against matched random-direction nulls and reach controls. It is confirmatory: exploratory claims are registered and tested once on a sealed split. Applied to seven TSFMs (TimesFM, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5, Timer and Time-MoE), it finds that causal features mainly change forecast level and seasonality; that models agree on which inputs a concept groups (20 of 20 registered claims confirmed on 971 held-out series) but that no cross-model claim about what a concept does confirms there (multi-model concepts 0 of 3, shared-input agreement 0 of 6 tested); and that no concept spans more than three of the seven models. In all, 43 of 82 replicable registered claims confirm: models share what they select, not demonstrably what they do with it. A planted-concept forecaster showed our first ablation null to be lenient, and we document fifteen plausible results that the system’s own controls showed to be wrong.

Keywords: mechanistic interpretability, time-series foundation models, sparse autoencoders, causal intervention, pre-registration

1 Introduction

Pretrained forecasting models are now default components. TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024), Moirai (Woo et al., 2024), MOMENT (Goswami et al., 2024), Lag-Llama (Rasul et al., 2023) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. Their architectures differ sharply (decoder-only stacks over 32-step patches, encoder-decoders over quantized single steps, encoders with cross-series attention, sampled flow-matching heads), and leaderboards say which is more accurate. They do not say what these models represent, whether architectures this different represent the same things, or whether any internal structure we can name drives the forecast.

Mechanistic interpretability offers probes (Alain and Bengio, 2017; Hewitt and Liang, 2019), representational similarity (Kornblith et al., 2019), activation patching (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024), lenses (Belrose et al., 2023) and sparse dictionaries (Bricken et al., 2023; Gao et al., 2025), packaged in libraries such as TransformerLens (Nanda

and Bloom, 2022) and pyvene (Wu et al., 2024), and early work applies pieces of this to TSFMs (Wiliński et al., 2025; Mishra, 2026; Oublal et al., 2026; Divo et al., 2026). Three properties of forecasting models break the direct transfer, and each causes silent errors rather than crashes:

- (i) **Tokens are not comparable across models.** One token is 1 time step in Chronos-T5, 32 in TimesFM, 16 in Chronos-2 and 96 in Timer, so every cross-model comparison needs a verified mapping from tokens to time.
- (ii) **There is no ground truth for concepts.** Real series carry no labels for trend, period, changepoints or noise process, so a probe that “finds seasonality” cannot be checked.
- (iii) **Effects are small and easy to fake.** Removing one sparse feature typically moves a forecast by a few hundredths of the context’s standard deviation, less than the dictionary’s own reconstruction error. At that scale a precision mismatch, a patch the head never reads or a null measured on the wrong rows yields a clean, confident and wrong result.

We present TSFM-LENS, a system designed around these problems, and use it to study three questions: **RQ1**, what forecast-relevant concepts TSFMs learn; **RQ2**, whether these concepts are shared across architectures, in the sense of responding to the same inputs, having the same effect, or both; and **RQ3**, whether they causally affect the forecast and how much of it they account for. A supporting question, **RQ4**, asks where in depth these computations happen and how comparable depth is across architectures.

TSFM-LENS rests on three commitments. It is *controlled*: a synthetic benchmark with exact generative ground truth and a battery of structured perturbations gives every probe a known answer. It is *causal*: a feature counts only if removing it changes the forecast more than removing a random direction of the same size on the same series, with reach controls proving the intervention landed. It is *confirmatory*: dev findings are frozen in a hash-pinned registry and tested once on a sealed private split. Measured *time alignment* makes architectures comparable, and a *one-command report* explains every figure in plain language.

Contributions.

- C1.** An open-source system (~61k lines, 171 test modules) that turns a configuration listing checkpoints into a self-contained HTML report through 19 fingerprinted stages, with a zero-code adapter path (Section 4).
- C2.** A controlled benchmark generator with two leakage tiers, a leakage audit, sealed splits with fresh epochs and bit-exact regeneration (Section 4.1).
- C3.** A causal concept pipeline that separates two notions of sharing that are usually conflated: *same inputs* and *same effect on the same inputs* (Section 5.3).
- C4.** A statistical protocol with series-level resampling, a multiplicity ledger and one-shot confirmation (Section 6).

C5. A seven-model study answering RQ1–RQ4 with typed evidence and held-out replication (Section 7), and a catalogue of fifteen cases in which an uncontrolled analysis gave a plausible, wrong answer that a built-in control caught (Section 8).

Main findings. On TimesFM 2.5, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5-Base, Timer and Time-MoE: (RQ1) causal sparse features exist (194 across 55 measured layers clear a profile-matched null, 2.47 times the empirical chance rate, though Timer’s do not exceed chance) but act mostly on forecast *level* and *seasonality*, and tile a continuum rather than forming discrete concepts; (RQ2) models agree on *which inputs* a concept groups (20 of 20 registered transfer claims confirmed on 971 sealed series) but we cannot show that they agree on *what it does* to them: no cross-model causal claim confirms held-out (multi-model atlas concepts 0 of 3, shared-input agreement 0 of 6 tested with 5 not testable), the atlas is segregated by model, and no concept spans more than three of the seven models; (RQ3) single features are causal but small and fragile (10 of 31 registered claims confirm on private data), and 84% of prominent features have no causal effect; (RQ4) depth-resolved reaction profiles largely replicate held-out (6 of 9 corruptions), but depth locations are comparable only after accounting for captured compute (14% to 99.8% of forecast FLOPs). Of 82 replicable registered claims, 43 confirm on the one-shot private split; the project’s reading is that models share what they select, not (demonstrably) what they do with it.

The full ledger of results, each with exact numbers, evidence class, status and a reproduction pointer, is `FINDINGS.md` (indexed in Appendix A). The example run of Section 7, with its report and every per-stage artifact, is `examples/panel7_v2; paper/figures/make_figures.py` regenerates every data figure from it.

2 Related Work

TSFMs. TimesFM (Das et al., 2024) is decoder-only over 32-step patches; Chronos (Ansari et al., 2024) quantizes values and trains a T5 encoder–decoder (Raffel et al., 2020) with a sampled decoder; Chronos-Bolt uses a direct patch head, and Chronos-2 (Ansari et al., 2025) is an encoder with cross-series group attention; Sundial (Liu et al., 2025) samples from a flow-matching head (Lipman et al., 2023); Timer (Liu et al., 2024) uses 96-step tokens, Time-MoE (Shi et al., 2025) a sparse mixture of experts over per-step tokens, and Lag-Llama (Rasul et al., 2023) reads lagged values. Most normalize each series internally (Kim et al., 2022), a detail that matters when an adapter calls a model below its public API (Section 8). Wan et al. (2026) document flat “collapsed” forecasts on weakly predictable targets, which we also measure (PM-07).

Interpreting time-series models. Saliency methods for time-series classifiers are unreliable across architectures (Ismail et al., 2020); Kalnāre et al. (2025) apply patching and SAEs to time-series classifiers. For TSFMs, Wiliński et al. (2025) study representational redundancy and concept steering, and four recent studies train SAEs on one TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026). Each examines one model or one method. None compares architectures on a common time axis, validates probes against generative ground truth, separates input agreement from effect agreement, or tests findings on held-out data.

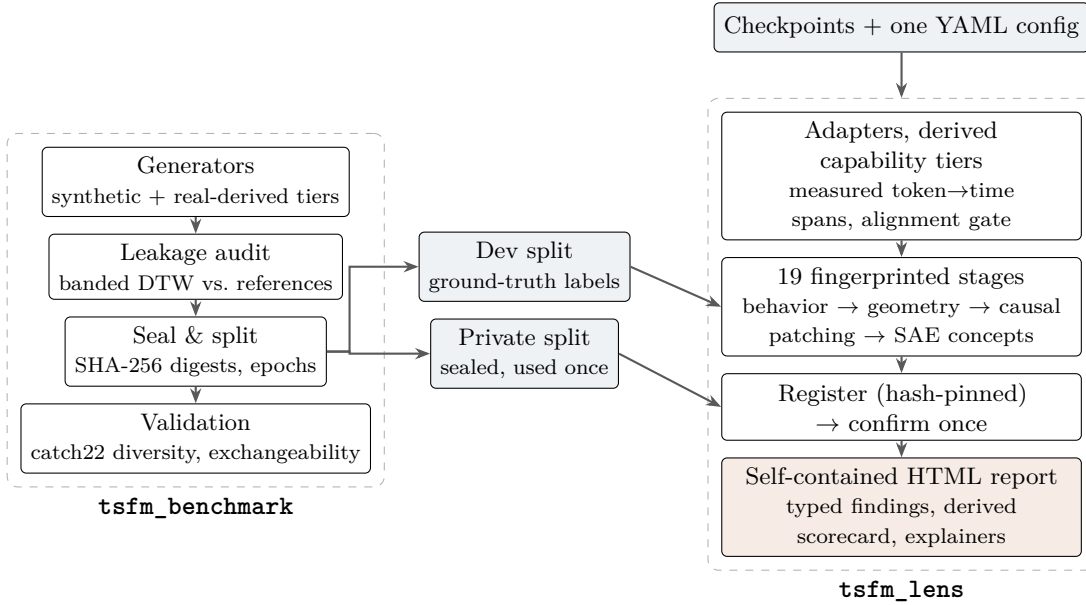


Figure 1: System overview. **tsfm_benchmark** produces leakage-audited, sealed corpora with exact ground truth; **tsfm_lens** turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by **confirm**, for claims registered in advance.

Methods, tooling and validity. Probes need control tasks (Hewitt and Liang, 2019; Belinkov, 2022); patching is sensitive to metrics and corruptions (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024); attention weights are not explanations (Jain and Wallace, 2019); SAE features vary across seeds (Paulo and Belrose, 2025). Representation similarity (Raghu et al., 2017; Kornblith et al., 2019; Bansal et al., 2021) underlies claims of convergence (Li et al., 2016; Huh et al., 2024). Libraries such as TransformerLens, nnsight (Fiotto-Kaufman et al., 2025), pyvene, SAELens (Bloom et al., 2024) and Captum (Kokhlikyan et al., 2020) provide primitives for one model, not comparative designs, nulls, multiplicity control or confirmation (Table 1). From empirical science we import pre-registration (Nosek et al., 2018), protection against adaptive reuse of holdouts (Dwork et al., 2015), leakage auditing (Kapoor and Narayanan, 2023) and series-level resampling (Efron and Tibshirani, 1993; Cameron et al., 2008).

3 Design Principles

Each principle corresponds to a failure that actually happened (Section 8) and is enforced in code as a gate, refusal or test.

P1. Ground truth by construction. Every probe is first validated on a case whose answer is known: the benchmark records every generative component of every synthetic series, and every perturbation is applied to series whose components are known.

Capability	<i>TransformerLens</i>	<i>msight / pyvene</i>	<i>SAELens</i>	<i>Captum</i>	<i>TSFM SAE studies[†]</i>	<i>TSFM-Lens</i>
Hooks, caching, activation patching	✓	✓	○	○	○	✓
Sparse dictionaries and feature ablation	○	○	✓	✗	✓	✓
Time-series models without per-model engineering	✗	○	✗	○	✗	✓
Measured token→time alignment across architectures	✗	✗	✗	✗	✗	✓
Benchmark with exact generative ground truth, leakage audit	✗	✗	✗	✗	○	✓
Random-direction nulls and reach controls for every ablation	✗	✗	✗	✗	✗	✓
Cross-model concept matching (inputs vs. effects)	✗	✗	✗	✗	✗	✓
Series-level resampling and multiplicity ledger	✗	✗	✗	✗	✗	✓
Registered, one-shot confirmation on sealed data	✗	✗	✗	✗	✗	✓
Typed evidence class and generated caveat per claim	✗	✗	✗	✗	✗	✓
One-command report designed for non-experts	✗	✗	○	✗	✗	✓

Table 1: Capabilities for comparative TSFM interpretability. ✓ provided; ○ possible with user code or partial; ✗ not provided. General libraries are single-model microscopes and complementary to TSFM-LENS, which uses the same primitives. [†]Mishra (2026); Oublal et al. (2026); Divo et al. (2026); Yıldırım (2026), judged from the papers.

P2. Alignment is measured, not declared. Token spans are measured from impulse responses; cross-model analysis happens only on a common window axis; models that are not time-localized receive behavioral analysis only.

P3. Causal claims need a null, a reach control and a same-seed baseline. Every intervention is compared with a random intervention of matched size on the same series; self-patching must change the forecast by exactly 0.0 and cross-patching by a nonzero amount; sampled models are seeded identically in compared calls.

P4. The series is the unit of inference. All intervals and p -values resample series; p -values are floored at $1/n_{\text{boot}}$, and each correction family’s attainable minimum is checked.

P5. Explore, then confirm once. Dev results are exploratory; registered claims are tested once on a sealed split, and a second look is recorded.

P6. Every claim is typed; failures are loud. Each finding carries an evidence class (BEHAVIORAL, DESCRIPTIVE, GEOMETRIC, LINEAR-TRANSLATABLE, CAUSAL WITHIN-MODEL, ILLUSTRATIVE, HELD-OUT CONFIRMED) and a generated caveat; unsupported analyses are skipped with a rendered reason.

P7. Written for newcomers. The deliverable is one HTML file, designed with the intention that a reader new to interpretability can follow it, with every figure explained and every verdict derived from artifacts rather than written by hand.

New behavior is opt-in so sealed corpora regenerate bit-exactly, and stage configurations are fingerprinted so stale cached artifacts are refused.

4 System Architecture

The repository holds two separately installable packages: `tsfm_benchmark` (one subpackage for generation, audit and sealing, another for diversity and exchangeability) and `tsfm_lens` (subpackages for models, extraction, analysis, SAEs and the report, plus the stage orchestrator `pipeline.py`). The folder `examples/panel17_v2` contains the report, configuration and every lightweight artifact of the run analyzed in Section 7, so the report and this paper’s figures can be regenerated without GPUs. The `main` branch is the clean publishable surface; a `dev` branch keeps the research log, standalone study drivers and experiment configurations.

4.1 Controlled benchmark generation

Real data enter the benchmark only as leakage references, as realism calibration and as raw material for explicitly labeled generators. The *synthetic* tier touches no real data and has exact ground truth: **parametric** is additive (trend, multiple seasonalities, AR noise, change-points, anomalies, random walk, intermittency, heteroskedasticity), and **random_parametric** draws a recipe from one of eight frozen archetypes (four more are opt-in). The *real-derived* tier (**mixture**, **block_bootstrap** (Künsch, 1989), **sequential_par**) inherits its source distribution and has attributional ground truth. Because “no leakage” and “realistic” pull in opposite directions, each sample carries its tier label and analyses report per tier.

Every sample passes a two-stage leakage gate against the Monash archive (Godaheewa et al., 2021) (a cheap prefilter, then banded DTW; Sakoe and Chiba, 1978) and a cross-split near-duplicate check. Dev and private splits use disjoint seed ranges mixed with SHA-256; every sample is hashed so that `load_sealed(verify=True)` detects tampering, and fresh private *epochs* are minted by striding the seed space. A separate validation package measures redundancy by shape matching and diversity in the catch22 feature space (Lubba et al., 2019) (participation-ratio dimension, Gao et al., 2017; nearest-neighbor tails), never on a 2-D embedding, and tests dev/private exchangeability with an energy distance (Székely and Rizzo, 2013) plus TOST equivalence (Schuirmann, 1987) (v1 benchmark: energy $p = 0.51$, 21 of 22 features equivalent; MP-09).

Two perturbation families are built in: nine structured *corruptions* (noise, detrend, deseasonalize, frequency shift, level shift, spike, smooth, warp, dropout), which drive L3, and *generator counterfactuals*, which move one generative knob along a dose ladder with every random draw fixed. Dose 1.0 must reproduce the original bit-exactly, a free identity test that caught a real bug (BM-05).

4.2 Model adapters and capability tiers

A model enters through an adapter (`tsfm_lens/tsfm_lens/models/base.py`) exposing `load`, `predict` (point forecast and quantiles in the original scale), the module, input preparation, a forward pass and `token_time_spans`. The *capability tier* is derived from which methods a subclass overrides, never declared: 0 (black box), 1 (observable hidden states with spans), 2 (steerable: single-pass context, so patching is possible) and 3 (decomposable attention). Stages above a model’s tier are skipped with the missing method named in the report. A missing capability (`CapabilityUnavailable`) is a fact about the adapter; a

Model	Checkpoint	Architecture	Token	Head	Tier	Params	FLOPs obs.
TimesFM 2.5	google/timesfm-2.5-200m	decoder-only	32	deterministic	3	231M	42.5%
Chronos-2	amazon/chronos-2	encoder (+group attn.)	16	deterministic	3	119M	97.5%
Sundial	thuml/sundial-base-128m	decoder + flow head	16	sampled	2	128M	21.3%
Chronos-Bolt	amazon/chronos-bolt-small	encoder + patch head	16	deterministic	2	48M	79.1%
Chronos-T5-Base	amazon/chronos-t5-base	encoder-decoder	1	sampled	3	201M	13.9% ^a
Timer	thuml/timer-base-84m	decoder-only	96	deterministic	2 ^b	84M	99.8%
Time-MoE	Maple728/TimeMoE-50M	decoder-only, MoE	1	deterministic	2 ^b	113M	50.0%
Lag-Llama	Lag-Llama	decoder over lags	lags	sampled	2	–	–

Table 2: Models with adapters; the first seven form the case study (Chronos-T5 also comes in other sizes, 8M–709M). “Token” is time steps per token; “FLOPs obs.” is the share of a forecast’s FLOPs spent in captured blocks at context 480, measured with PyTorch’s FLOP counter. ^aEncoder only (DE-05). ^bVia `generic_hf`.

measured failure of time localization (`NotTimeLocalized`) is a fact about the model and routes it to behavioral analysis only.

New models use the zero-code `generic_hf` adapter, which discovers layers and measures spans and refuses rather than guesses (its span discovery reproduces all six hand-written adapters’ spans exactly, mean IoU 1.0, and recovers Timer’s 96-step spans from measurement alone; MP-01), or a copy of `tsfm_lens/tsfm_lens/models/TEMPLATE_adapter.py` in `models/contrib/`. `run.py --check-adapter` then runs patch-identity and reach conformance, the alignment gate, span discovery and capability checks. Table 2 lists the models.

4.3 Time alignment across architectures

Each token’s span is measured from its impulse response (`extraction/span_discovery.py`), gated on peak-to-pedestal contrast and contiguity. Given spans I_j and windows $W_w = [wL, (w + 1)L)$ ($L = 32$ by default), tokens are overlap-pooled,

$$P_{wj} = \frac{|I_j \cap W_w|}{\sum_{j'} |I_{j'} \cap W_w|}, \quad \tilde{h}_w = \sum_j P_{wj} h_j, \quad (1)$$

and every cross-model analysis runs on [series, window, dim] tensors (Figure 2); an empty window raises rather than yielding a zero row. During extraction an *alignment gate* injects an impulse into each window and checks, per layer, that the matching pooled window moves most, normalizing hits by their attainable ceiling (a 96-step token on 32-step windows reaches at most one third; MP-02), with the impulse amplitude self-calibrated per checkpoint and context length.

4.4 The stage pipeline and its gates

An analysis is a pipeline of 19 stages (Figure 3) writing typed artifacts. Stages skip themselves when their artifacts exist; each declares the configuration keys it depends on, which are fingerprinted so that a changed setting marks exactly the dependent artifacts stale. Three gates write their decisions to JSON and the report names them as skip reasons: the *tier* gate, the *run-shape* gate (comparison stages are dropped for one model) and the *routing* gate.

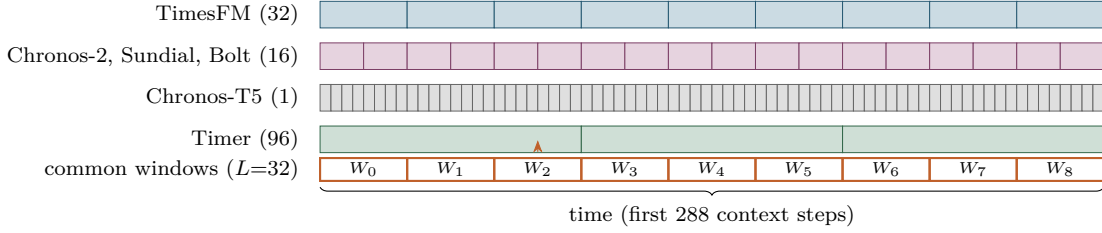


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5’s one-step tokens drawn in groups of four); Equation (1) pools them onto common windows.

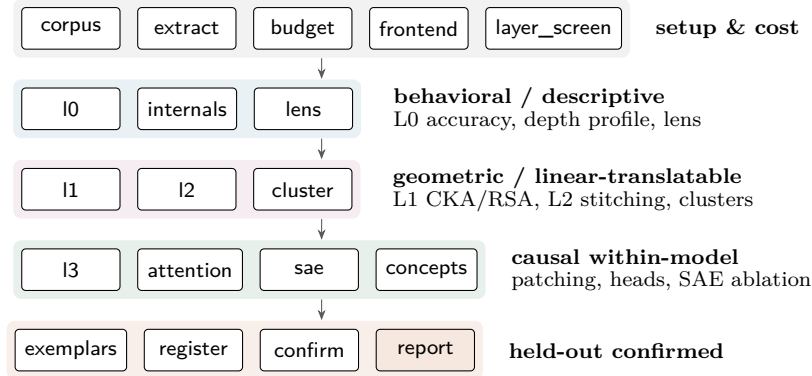


Figure 3: The 19-stage pipeline, grouped by the strongest evidence class each group can produce. Gates (tier, run shape, routing, configuration fingerprints) skip a stage per model with a rendered reason; only **confirm**, on the sealed private split, produces HELD-OUT CONFIRMED evidence.

Extraction captures the residual stream into a float16 zarr store, and every intervention goes through one primitive (`hooks.token_patch`), which raises if the model silently splits a batch.

4.5 The report: designed to be readable by newcomers

The deliverable is one self-contained HTML file (`examples/panel17_v2/report.html`), designed with the intention that a reader without interpretability training can follow it, while an expert can audit every number (Figure 4).

- **Question-first.** Seven parts pose plain questions (“Is the comparison fair?”, “Where does the forecast form?”, “What causally drives the forecast?”, ...); each stage opens with a card stating its question, method, what good and bad results look like, and what it cannot tell you.
- **Every figure explained.** A caption plus a collapsible “What does this mean?” note (purpose, how to read, limitations): 530 notes in the example. A glossary and methods appendix are generated from the code.

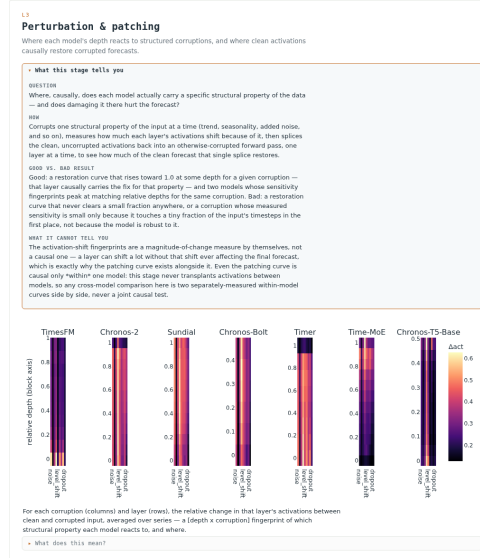


Figure 4: Excerpt of the generated report (L3 section of the seven-model example report, `examples/panel7_v2`): the stage card states the question, method, how to read good and bad results and the limitation, above the per-model corruption-sensitivity fingerprints. Every figure carries a caption and an expandable explanation.

- **Derived verdicts.** The scorecard is a pure reduction over artifacts with no model names hard-coded; each row shows the value, its reference (null, floor, ceiling or baseline) and the rule that yields the verdict.
- **Typed findings.** Each claim is a **Finding** record with evidence class, plain-language sentence and generated caveat: 240 in the example (89 descriptive, 54 behavioral, 49 causal, 42 linear-translatable, 4 geometric, 2 illustrative).
- **Nothing silently missing.** Skipped stages render their gate reason, failed sections render visibly, and a multiplicity ledger (123 families, 4,238 comparisons) flags families that cannot reach α .

The complete analysis is one command,

```
python run.py --config configs/examples/large_4model.yaml
```

and smaller presets cover a CPU-only mock smoke test, a single model and a pair.

5 Measurement Methods

Throughout, $x \in \mathbb{R}^T$ is a context ($T = 480$ in the case study), $y \in \mathbb{R}^H$ its continuation ($H = 64$), $\hat{y}$ a point forecast and $h^{(\ell)}$ the residual stream after block ℓ .

5.1 Behavior, geometry and perturbation (L0–L3)

L0. Accuracy is MASE (Hyndman and Koehler, 2006) per series, with the seasonal lag from ground truth and a scale floor on the naive denominator, without which intermittent

series spuriously improve (MN-23); sMAPE, pinball loss (Gneiting and Raftery, 2007) and calibration are reported alongside. Pairs are compared by a paired series bootstrap per family, Holm-corrected across all pairs and families (Holm, 1979), and read against each model’s repeat-run noise floor. **budget** measures parameters, FLOPs, latency and captured-FLOP share; **frontend** tests scale equivariance, NaN handling and quantization before any layer runs.

L1, L2. For aligned, window-pooled and centered activations A, B over the same series, linear CKA (Kornblith et al., 2019) $\|B^\top A\|_F^2 / (\|A^\top A\|_F \|B^\top B\|_F)$ is computed for every layer pair of every model pair, with a cluster bootstrap and a shuffled-series null. L2 fits ridge maps between layers and reports only the *gain* over an input-feature baseline, since much of any layer is linearly predictable from the input. Cluster agreement uses AMI (Vinh et al., 2010). These claims are GEOMETRIC or LINEAR-TRANSLATABLE.

L3. For each corruption c , the sensitivity *fingerprint* is the relative activation change $\|h^{(\ell)}(c(x)) - h^{(\ell)}(x)\| / \|h^{(\ell)}(x)\|$ per layer, and the *restoration curve* is the fraction of forecast damage undone by patching clean activations of layer ℓ (whole, or per window) into the corrupted pass. Models are compared by rank-correlating fingerprints on shared relative depth, never by transplanting activations. Restoration is CAUSAL WITHIN-MODEL; cross-model agreement is correlational.

5.2 Depth

A skip lens decodes the forecast from each layer through the model’s head and a tuned ridge lens (Belrose et al., 2023) fits per-layer readouts; **internals** reports effective dimension, family decodability and CKA to the input. The depth axis is the fraction of the *whole* stack, including uncaptured parts, so Chronos-T5’s last captured block sits at 0.478 (DE-04); and the lens is withheld for a model whose head reads positions the patch never writes, where it would return a flat, meaningless curve (DE-03).

5.3 Causal concepts

Sparse dictionaries. Each target layer gets a TopK SAE (Makhzani and Frey, 2013; Gao et al., 2025), $z(h) = \text{TopK}(W_e(h - b_d) + b_e)$, $\hat{h} = W_d z(h) + b_d$, with searched dictionary size, an auxiliary loss reviving dead atoms and a minimum step count (MP-03). Since single seeds are not reproducible (Paulo and Belrose, 2025), replicate dictionaries give a within-model reproducibility ceiling (MP-04).

The ablation battery. Feature i is tested on the k series where it fires most, by comparing the forecast under the reconstruction with $z_i = 0$ against the full reconstruction:

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \quad (2)$$

where g_c is one of nine forecast channels (trend, seasonal strength, spectral centroid, level, dispersion, near/far-horizon shape distance, MASE, flatness). Using the reconstruction as baseline keeps the SAE’s error from being charged to the feature. The null is *profile-matched*: on the *same* rows, each of 16 draws removes, at every token t , exactly the amount the feature removes, $|z_i(t)| \|w_i\|$, along a random decoded direction, so only the direction differs. A

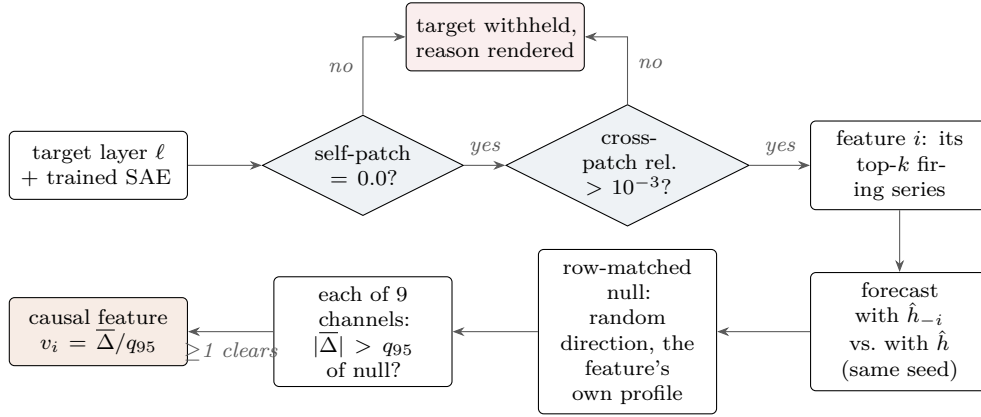


Figure 5: The ablation battery. A target is scored only after both reach controls pass; each feature is compared, channel by channel, with a random-direction null on exactly its own top-firing series.

feature clears channel c if mean $|\Delta_i^{(c)}|$ exceeds the pooled null’s 95th percentile $q_{95}^{(c)}$, and its channel vector is $v_i^{(c)} = \overline{\Delta_i^{(c)}} / q_{95}^{(c)}$. Shape channels are distances and are treated as undirected. Because a mean over rows is compared with a percentile of single rows, the chance rate is not 5%; it is estimated per cell by scoring each null draw against the others (0.3–4% per target on the case study). An earlier null removed one uniform, batch-averaged amount; the known-answer test of Section 5.5 showed it lenient (MN-29). A *reach probe* first requires self-patching to change the forecast by exactly 0.0 and cross-layer patching by a relative $> 10^{-3}$; otherwise the target is withheld with its reason (Figure 5).

From features to concepts. Causal features are grouped at three granularities. *Per-target concepts* cluster one layer’s vectors with k -means, choosing k by silhouette (Rousseeuw, 1987) among values whose smallest cluster has ≥ 3 members (without it, the four highest silhouettes on the earlier four-model run, 0.607–0.717, were all one-outlier splits; SH-20); a target with no admissible k is *non-modular*. The *atlas* pools features from all models and layers and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a shared causal-effect profile. *Families* are a coarser readable tiling (average linkage, cosine 0.5), tested against a column-shuffle null and titled deterministically from their channels.

5.4 Two notions of sharing

“Shared” is two questions, answered separately. *Same inputs*: concept transfer asks whether a feature in another model separates a concept’s top- k series from the rest (AUC), against stratum-matched random subsets, in both directions (*reciprocal* if both clear), with Benjamini-Hochberg (Benjamini and Hochberg, 1995) per ordered model pair and leg. Input agreement between concept parts is top- k overlap (hypergeometric), never whole-series correlation: two sparse features can have $\rho = 0.80$ with no shared top series (MN-19). *Same effect on the same inputs*: every reciprocal-FDR transfer is ablated on both sides on the same series set U ; each side must clear its own null on U , and level concordance and shape cosine must

beat both sides’ activation-matched random-feature floors. Falling below the floors’ 5th percentile is “acts differently”; merely failing the 95th is “no specific agreement” (MN-18).

The report combines these into a six-rung ladder per concept (`report/derived.py`): L1 same forecast effect (atlas), L2 same inputs, L3 reproducible across SAE seeds, L4 other dictionaries select the same inputs, L5 same causal effect on the same inputs, L6 confirmed on private data. Each atlas concept also gets a `sharing_class`: *shared* (same effect and inputs), *partially shared*, *convergent* (same effect, different inputs) or *single-model*; a part is provenance-driven when $\geq 80\%$ of its top series come from one real-derived generator.

5.5 Known-answer validation

Every causal count in Section 7 rests on a null, and a null can be wrong without any error being raised. TSFM-LENS therefore ships a forecaster whose answer is known, `mock_planted`. One planted block writes known unit directions into the residual stream on the series that have a property (thresholded readouts of the patch mean, slope, residual deviation, seasonal projections and a spike statistic, each firing on 10–30% of the corpus), and the head applies a known effect (level, trend, seasonality or dispersion) to the forecast. It carries decoys written like concepts (*input-only*, zero causal effect; *sub-null*, weak effect), a control layer after the planted block with no planted effect, and, for the cross-model rungs, a pair of architectures (widths 64 and 96, depths 5 and 7, rotated residual basis) sharing planted concepts plus an opposite-effect twin. A scorer reads only the planted manifest and the run’s artifacts, against a gate fixed before any data was seen: per-cell false-positive rate on the control layer at most 0.10 and sensitivity at least 0.5, on five construction seeds, seed 0 used to tune the attribution rule and seeds 1–4 held out. The gate is evaluated at one effect size (dose 1).

The legacy null fails the gate on all five seeds (per-cell false-positive rate 0.168–0.248): it is sized to a chunk-wide mean $|z|$, zeros included, and removed uniformly, whereas a feature’s ablation removes its full activation where it fires hardest, so a strong or dense atom clears it on size alone (17–25% of the cells and 34–47% of the features of a layer with no planted effect cleared it). The profile-matched null passes on all five seeds (per-cell false-positive rate at most 0.003, sensitivity 0.70–1.00; a random decoded direction overlaps the planted span more than a decoy does, so this null may be over-strict in the planted world); on seed 0 it also stops the legacy null’s 4 of 4 sub-null decoys from clearing, lifts the atlas adjusted Rand index from 0.336 to 0.469 and raises the sharing-class accuracy from 0.364 to 0.714. Re-scoring the four-model case study under it cut the causal features from 293 to 110 and the convergent concepts from 13 to 0, which is why the earlier numbers are upper bounds. The same device was applied to the cross-model agreement rung of Section 5.4: it never calls a decoy “same” (0 of 9), but finds a truly shared single-direction concept in only 0.4 of seeds and one spread over 3–4 directions in 0.0 (MN-30), and an opt-in rule that requires a statistic to be defined before it can fire removes its false disagreements on a shared concept (3 of 30 to 0 of 30). Known-answer tests establish what a method can detect, not that it is right on real models.

6 Statistical Protocol

Four mechanisms control how often a report with thousands of comparisons is fooled (Table 3).

The series is the resampling unit. Windows of a series share its trend, noise and scale, so every interval and p -value resamples whole series (Efron and Tibshirani, 1993; Cameron et al., 2008): paired bootstrap for accuracy, cluster bootstrap for CKA, validation-series intervals for stitching, paired cluster bootstrap for patching. Splits are by series and stratified, never head slices of a task-grouped corpus.

A multiplicity ledger. Every test belongs to a named family (Holm for confirmatory families, BH for discovery-oriented atlas transfers). Resampled p -values are floored at $1/n_{\text{boot}}$ (Phipson and Smyth, 2010), so a Holm family of m tests cannot go below m/n_{boot} ; the ledger computes this minimum per family and flags families that can never be significant, a failure that grows quadratically with the number of models. The example has 123 families and 4,238 comparisons; its strictest L0 family (105 tests over 21 pairs) has a minimum Holm p of 0.021 at $n_{\text{boot}} = 5000$.

References from the run itself. Effects are read against repeat-run noise floors, shuffled-series nulls, input-feature baselines, random-direction nulls, activation-matched floors and replicate-seed ceilings. Before the private split is opened, `confirm` simulates each claim’s minimum detectable effect (e.g. ΔMASE 0.133 at $n = 290$, 80% power), so a non-replication reads as “absent” or “underpowered”.

Register, then confirm once. `register` freezes hypotheses from dev results together with the content hash of each source artifact, and hashes the registry. `confirm` refuses mismatched hashes, requires the private seal to verify, tests each claim once and refuses to overwrite its artifact; a forced rerun is recorded as `repeated_look`, and a consumed split is replaced by a fresh epoch, checked for sample-hash overlap with every earlier split of every corpus version (an epoch minted for the case study was rejected unopened because 716 of its 961 series regenerated a split an earlier confirmation had consumed). This is pre-registration (Nosek et al., 2018) in software, guarding against adaptive holdout reuse (Dwork et al., 2015); unregistered findings cannot be rescued.

7 Case Study: Seven Time-Series Foundation Models

7.1 Setup

We analyze seven models with the preset `tsfm_lens/configs/panel7_v2.yaml`: TimesFM 2.5 (Das et al., 2024), Chronos-2 (Ansari et al., 2025), Sundial (Liu et al., 2025), Chronos-Bolt and Chronos-T5-Base (Ansari et al., 2024), Timer (Liu et al., 2024) and Time-MoE (Shi et al., 2025); the last two load through the zero-code `generic_hf` adapter. The dev corpus has 965 series in five generator families; the real-derived ones draw on Monash weather, ETT (Zhou et al., 2021) and the Chronos electricity and traffic collections. Contexts are 480 steps (a multiple of Timer’s 96-step token), the horizon 64, the window 32. SAEs are trained on 56 target layers (TimesFM 10, Chronos-2 9, Sundial 9, Chronos-Bolt 4, Timer 6, Time-MoE 9, Chronos-T5-Base 9) with three replicate seeds each, and $n_{\text{boot}} = 5000$. The private split is 971 sealed series (v2 epoch 3), opened once; its provenance is in Appendix C. The run recorded 21.1 hours of stage time on one RTX A5000 (24 GB). Artifacts are in `examples/panel7_v2/run`. Ledger-tagged numbers measured

Claim type	Test statistic	Null / reference	Correction	Held-out
Model A more accurate (per family)	paired Δ MASE, series bootstrap	zero difference; repeat-run noise floor	Holm over all pairs $\times$ families	✓
Layers share geometry	linear CKA, cluster bootstrap	shuffled-series null at the same layer pair	per model pair	✓
Layer linearly translatable	ridge stitching gain	input-feature baseline (gain = 0)	per pair	–
Depth reaction agrees	Spearman ρ of fingerprints on shared depth	bootstrap CI of ρ	per corruption	✓
Layer restores forecast	restoration fraction	self-patch = 0, cross-patch $\neq$ 0	per model	–
Feature is causal	mean $ \Delta^{(c)} $ on own rows	profile-matched random-direction q_{95}	per channel (count vs. chance); Holm when registered	✓
Concept reproducible	best-match cosine across seeds	within-model replicate ceiling	–	–
Same inputs	top- k AUC in other model	stratum-matched random subsets	BH per ordered pair and leg; Holm when registered	✓
Same effect on same inputs	level concordance; shape cosine	activation-matched feature-set floors (q_{95} agree, q_{05} differ)	per test; Holm when registered	✓
Concepts are cross-model	atlas centroid cosine and member pairs	column-shuffle and model-label nulls	Holm when registered	✓
Internals predict failure	gain in log-MASE Spearman over a free baseline	zero gain	Holm when registered	✓
Families are real	number and coverage of families	column-shuffle null	–	–

Table 3: Every claim type the case study reports, the statistic that tests it, the reference it must beat, its multiplicity correction and whether the **confirm** stage currently re-tests it on private data. Claim types without a held-out mark are reported as exploratory.

in another run say so (the earlier four-model run is `examples/concept_atlas_v2`); every other number is from this run or its confirmation.

Table 4 gives the models’ behavior. Chronos-2 is the most accurate; Time-MoE is less accurate than the naive forecast and Timer close to the seasonal naive; Sundial’s and Chronos-T5-Base’s nominal 80% intervals cover 36% and 44% of outcomes; and TimesFM, Sundial, Timer and Time-MoE propagate an injected NaN silently. The Sundial row hides a bug found in this work: the first adapter bypassed the checkpoint’s input normalization, and a scale-equivariance residual of 38.43 context standard deviations was read as a model property (PM-15, Appendix D; after the fix, 0.031).

7.2 RQ1: What forecast-relevant concepts do TSFMs learn?

Causal features exist, but not equally in every model, and the battery discriminates. 194 SAE features across 55 measured targets clear the profile-matched null on at least one channel. Fifty-five of 56 targets passed both reach controls (e.g. TimesFM `stacked_xf.10`: self-patch 0.0, cross-patch relative reach 0.253); Chronos-2’s final block was withheld because cross-patching moved the forecast by exactly 0 (its head reads placeholder tokens a context patch never reaches; DE-03). Against the leave-one-draw-out chance rate, 442 (feature, channel) cells clear the null where 178.75 are expected (2.47 times chance), from 1.02 times (Timer) to 5.70 times (Chronos-Bolt); 22 of 55 targets are individually

Model	median MASE	80% coverage	scale resid. (s.d.)	NaN input	latency (ms)
Chronos-2	1.2002	0.757	0.0102	handled	23
TimesFM	1.2560	0.779	1.6×10^{-6}	propagates	39
Sundial	1.4131	0.360	0.0310	propagates	62
Chronos-T5-Base	1.4231	0.436	2.7×10^{-7}	handled	975
Chronos-Bolt	1.5487	0.732	0.0101	handled	15
Timer	1.9306	–	0.0262	propagates	5
Time-MoE	2.8642	–	0.0368	propagates	103
naive / seasonal naive	2.7593 / 1.9819				

Table 4: Behavior on the v2 dev corpus (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations, $c = 0.001$. Latency: median time of a forecast for a batch of 8 series. Timer and Time-MoE output point forecasts only. Held-out, Chronos-2 beats TimesFM on the registered mixture family (paired ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$) and, with a smaller effect, overall (-0.070 [-0.129 , -0.008], $p = 0.025$, $n = 971$).

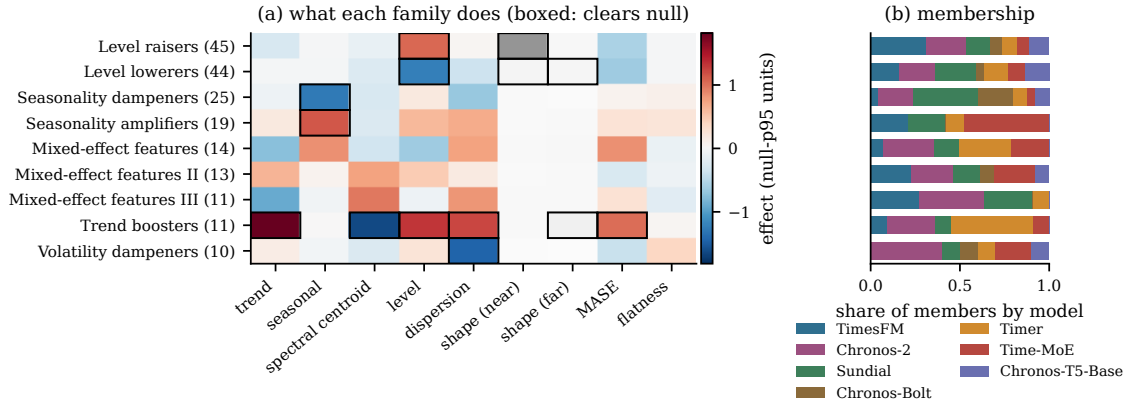


Figure 6: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature *does* (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

significant (BH at 0.05) and Timer has none, although its layers reach the forecast (relative reach 0.066–0.261; PM-17). (The nominal “ $0.05 \times \text{cells}$ ” line would have read it as 0.96 times chance; MN-29.)

The concepts are about level and seasonality. Nine effect families tile the causal features (Figure 6). The two largest act on level: *Level raisers* (45) and *Level lowerers* (44), 89 of the 192 assigned features. Seasonality dampeners and amplifiers follow (25 and 19); three mixed-effect families (14, 13, 11), trend boosters (11) and volatility dampeners (10) are smaller. Level is not everything: on the earlier four-model run (original null, an upper bound) the median pure-level share of an effect was 0.535 (CA-07).

The concepts are graded, not discrete. Within a layer, 50 of 56 targets are non-modular (10 per-target concepts in total, on 5 targets, 4 of them on Timer targets whose

clears are not above chance); the strict atlas assigns only 35 of 194 features to its 11 concepts, more than a column-shuffle null produces (null 95th percentile 5.05 concepts, assigned fraction 0.088 against 0.180; $p = 0.005$) but few; and the families cover 192 of 194 features but do *not* beat the shuffle null ($p = 0.68$ for their number, 0.23 for coverage). Silhouette is 0.25–0.32 for every cut from 6 to 16 families, and on the earlier run the ablation fingerprints had participation ratio 2.09 (CA-10, MN-07). Families are thus a reading vocabulary, not discovered units.

Ground-truth concepts are rare and need a null. After dead-feature revival, 154 of 10,240 TimesFM and 115 of 6,144 Chronos-T5-Base features match a ground-truth field beyond a permutation null (best: intermittency, $\rho = 0.881$; MP-06). Moving a generator knob was a confirmed negative twice (0 of 98, then 0 of 151 pairs; MN-17), and a vivid TimesFM “random walk” concept failed seed stability (PM-11).

Answer to RQ1. TSFMs contain sparse causal directions that mainly control forecast level and seasonality, in smaller groups trend, volatility and mixtures. They form a continuum with a few dominant axes, not discrete concepts aligned with named generative factors, and in one of seven models (Timer) the feature battery does not exceed chance on this corpus.

7.3 RQ2: Are concepts shared across architectures?

Geometry is shared above chance, but similarity has no single meaning. Peak CKA between TimesFM and each other model is 0.342–0.426 (shuffled-series nulls 0.011–0.051); the TimesFM/Chronos-2 peak replicates held-out (dev 0.4257; private 0.4109 [0.4014, 0.4266]), and the band holds across five architectures including Timer and Lag-Llama (SH-01). Yet seven pair metrics rank the 21 pairs only weakly consistently (Kendall’s $W = 0.25$, $p = 0.0095$; Figure 7): TimesFM/Chronos-2 is first by error agreement (0.948) and 15th of 21 by CKA, where Sundial/Chronos-Bolt is first (0.927), so the metrics are never pooled. Two other similarity measures fail their own controls (untrained pair more similar than a trained pair, MN-01; a unit-correspondence measure that rose under rotation, SH-19; Table 6, rows 11–12).

Models agree on which inputs a concept groups. 1,296 of 1,584 atlas transfer tests are reciprocal after BH correction. Twenty claims registered from dev alone were all confirmed on the 971 private series (Holm, maximum adjusted $p = 0.0100$, the floor for 20 claims at 2000 draws; AUC 0.919–0.970 against null 95th percentiles of 0.70–0.75; Figure 9b), replicating on four private draws (19/20, 20/20, 20/20 on earlier draws, and 20/20 here; SH-18, CA-11). This is an input-level claim. The 20 claims come from two concepts (TimesFM `xf.18`, 14 claims; Sundial `layers.6`, 6) and reach destinations in all seven models, with only 8 of 20 private best features identical to dev: the destination has *some* feature selecting the same series.

Models rarely agree on what a concept does, and the atlas is segregated by model. Of 11 atlas concepts, 9 have causal members from ≥ 2 models, but only 2 are *shared*, 2 partially shared, 5 *convergent* (same effect, different series) and 2 single-model (Figure 8); among the 9 seed-stable concepts the counts are 2, 1, 4 and 2. No concept spans more than 3 of the 7 models, and the clusters are *purier* by model than random assignment:

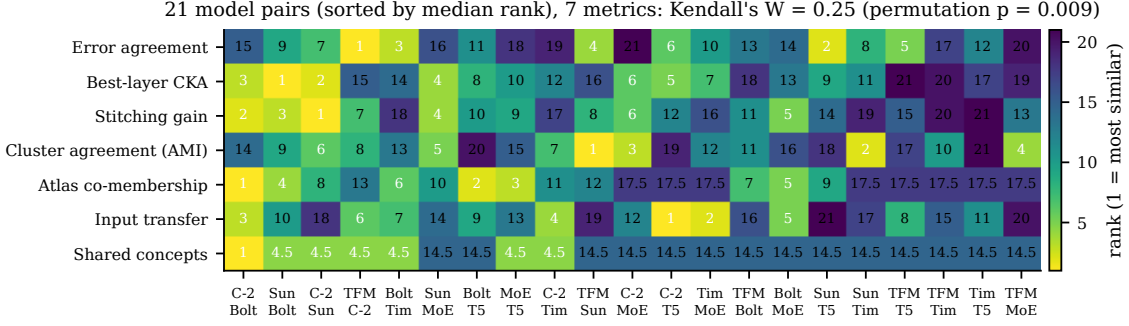


Figure 7: Rank of each model pair (columns, sorted by median rank; TFM TimesFM, C-2 Chronos-2, Sun Sundial, Bolt Chronos-Bolt, Tim Timer, MoE Time-MoE, T5 Chronos-T5-Base) under seven similarity metrics (rows); 1 is most similar. Source: `report/model_similarity.json`.

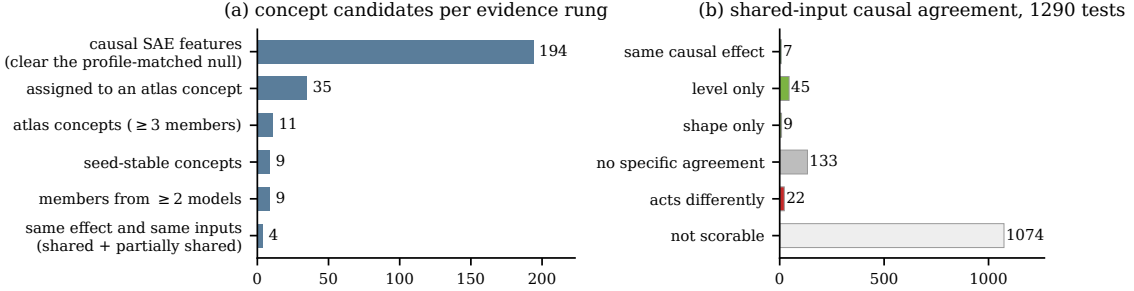


Figure 8: (a) Concept candidates at each rung (first three bars in sequence; last three are properties of the 11 atlas concepts). (b) Shared-input causal agreement over 1,290 tests drawn from the 1,296 reciprocal transfers. Source: `sae/concept_stage.json`.

9 multi-model concepts against a null mean of 10.8 ($p_{\text{below}} = 0.025$) and mean model purity 0.591 against 0.474 (SH-22). Of 1,296 reciprocal transfers, 1,290 were tested for shared-input agreement (a cap of 60 per ordered pair) and 216 are scorable; 7 (3.2%) show the same causal effect, 54 agree on level or shape alone, 133 show no specific agreement and 22 act differently. All 7 “same” verdicts are Chronos-2/Chronos-Bolt, six of them through one Chronos-Bolt block-4 feature set, so they are one concept seen from several layers. This pattern must *not* be read as sharing following lineage: 1,053 of the 1,074 unscorable tests fail because a side’s own ablation does not clear its own null on the roughly 24 shared series, and on a planted pair of different architectures the rung finds a truly shared single-direction concept in 0.4 of seeds, a concept spread over 3–4 directions in 0.0, with 0 of 9 false “same” verdicts (MN-30). The rung is specific but insensitive, so an absence of “same” across architectures is uninformative.

Answer to RQ2. Different TSFMs *select* the same series for a concept reliably enough to replicate on sealed data (four private draws), but we cannot show that they *use* them the same way: the atlas is segregated by model, no concept is common to the panel (a

structural claim that confirms held-out), the one clear same-effect case is inside the Chronos family, and no cross-model causal claim confirmed on private data (Section 7.7). Whole-layer similarity cannot reveal this and is internally inconsistent.

7.4 RQ3: Do concepts causally affect the forecast, and how much?

Individually, little, and not reliably. Of the 32 single-feature causal claims registered from dev, 31 were testable on the private split and 10 confirm (TimesFM 3 of 4, Time-MoE 2 of 7, Sundial 2 of 4, Chronos-Bolt 2 of 6, Chronos-2 1 of 7, Chronos-T5-Base 0 of 4; by channel, near-horizon shape 4 of 8, MASE 3 of 8, spectral centroid 2 of 4, dispersion 1 of 3, seasonal 0 of 4, level 0 of 4); each confirmed claim has $p = 0.001$ after an adaptive redraw to 1000 null draws and $p_{\text{Holm}} = 0.031$, the floor for 31 tests. Twenty-one do not confirm (19 of them were not at the 200-draw p floor), and one flips sign (Chronos-T5-Base block 10). Most prominent features carry nothing: 0.835 of the 927 prominent features of the private atlas are causally null (one-sided lower bound 0.817, dev 0.810, $p_{\text{Holm}} = 0.0005$; MN-14), and TSFM-LENS collapses them rather than rendering them. On the earlier four-model run the median single-feature effect was 0.0395 context standard deviations, 3.5 times below the 0.139 moved by substituting the SAE reconstruction (MN-15). Attention mass on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ TimesFM, -0.118 Chronos-T5-Base; CA-04; cf. Jain and Wallace 2019) (a minimal seasonality circuit is one head in TimesFM, five non-additive heads in Chronos-T5-Base; CA-03).

Answer to RQ3. Features and heads are causal in the controlled sense, but contributions are small and distributed, about one in three single-feature claims survives new series, most prominent features carry nothing, and attention is an unreliable guide.

7.5 RQ4: Where in depth, and how comparable is depth?

For TimesFM/Chronos-2, 6 of the 9 per-corruption fingerprint agreements replicate held-out (overall $\rho = 0.524$ [0.504, 0.544]; Figure 9a): the models react at matching depths to detrending (private 0.368), dropout (0.510) and, more strongly than on dev, frequency shift (0.381 against 0.216), and at opposite depths to noise (-0.743), spikes (-0.785), warps (-0.651) and smoothing (-0.581 against -0.418); deseasonalizing is near zero (0.086 [-0.027 , 0.186]). The three non-replications are the two stronger-on-private effects (frequency shift, smoothing) and level shift, whose sign flips (dev -0.188 , private 0.109 [-0.025 , 0.159]); under the registered rule, dev ρ inside the private interval, a stronger private effect also counts as a non-replication. In TimesFM, per-window restoration concentrates just before the horizon and grows with depth (earlier four-model run, **noise**: 0.068 at layer 0 to 0.412 at layer 16; exploratory, $n \approx 300$; DE-01). Depth *locations* need two qualifications that every report renders: captured compute ranges from 13.9% (Chronos-T5-Base) to 99.8% (Timer) of a forecast (DE-05), and Chronos-T5’s last encoder block sits at 0.478 on the whole-stack axis (DE-04). No selected layer was robust across three layer selectors for any model (DE-08).

Answer to RQ4. Depth-resolved reaction profiles are largely reproducible (6 of 9 corruptions); absolute depth locations are comparable only after accounting for captured compute and the depth convention.

7.6 Can looking inside predict forecast failure?

If internal features are causal, do they tell a user when to distrust a forecast, or which model to use? For each model we predict a series’ log MASE (Spearman) and failure flag (AUROC) from free output-only features (quantile width, flatness, catch22 of the context), then add the SAE-family activations at the last window (cross-fitted, series bootstrap, 941 dev series, $n_{\text{boot}} = 1000$; evidence class *predictive*, since a feature that predicts error is not a cause of it). The free baseline is strong (Spearman 0.76–0.85). For five of seven models (Chronos-2, TimesFM, Sundial, Chronos-T5-Base, Chronos-Bolt) the gain’s CI includes zero for both tasks (log-MASE Spearman gains -0.004 to $+0.005$, AUROC -0.010 to $+0.008$). It is positive for the two weakest models, Timer ($+0.030$ [0.019, 0.043], baseline 0.787, failure rate 0.30) and Time-MoE ($+0.012$ [0.005, 0.019], baseline 0.853, failure rate 0.62), robust across five fold seeds, and no failure-AUROC gain excludes zero. Both gains were registered and confirm when the frozen procedure is refit on the private split ($n = 935$): Timer 0.0446 [0.0292, 0.0605], $p_{\text{Holm}} = 0.002$; Time-MoE 0.011 [0.0032, 0.0194], $p_{\text{Holm}} = 0.012$ (14 dev comparisons were run and 2 selected, which is legitimate because the private split referees). Routing each series to the model with the lowest predicted MASE does not help: baseline routing reaches mean MASE 1.948 [1.814, 2.070], baseline plus internals 1.964 [1.837, 2.093] (gap -0.016 [-0.044 , 0.010], $p = 0.258$), and both are worse than always using Chronos-2 (1.856; $p = 0.001$; oracle 1.523). Internals add information where the free baseline is weakest and nowhere else.

7.7 What replicated on held-out data

One look at a fresh private corpus (v2 epoch 3: 971 series, no sample shared with any other split, seal verified) tested a registry of 128 hypotheses, 82 of them replicable. **43 confirm, 6 cannot be tested and 33 are not confirmed** (Figure 9; every Holm family is per claim type, as registered). What confirms: the accuracy claim (Chronos-2 better on *mixture*, ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$; minimum detectable effect 0.133), the peak-CKA pair, 6 of 9 perturbation agreements, all 20 transfer claims, 10 of 31 testable single-feature causal claims, the two structural negatives (no concept spans the panel: the private atlas has 10 concepts, at most 3 of 7 models each; most prominent features are causally null) and the two U1 claims. What does *not* confirm is every cross-model causal claim: the multi-model atlas concepts (0 of 3: their centroid cosines still beat the null, $p \leq 0.007$, but only one third of member pairs reach cosine 0.9; the one atlas claim that confirms is single-model, Chronos-2) and shared-input agreement (0 of 6 tested; 5 of 11 are not testable because a side’s feature does not clear its own null on the private shared series; the best, Chronos-Bolt block 4 to Chronos-2 block 7 “same”, reproduces its verdict with $p = 0.011$ but misses Holm at 0.066). Given the rung’s measured insensitivity (Section 7.3), the agreement failures are weak evidence of difference. The project’s held-out reading is that **models share what they select, not (demonstrably) what they do with it**. Two disclosures (the first is detailed in Appendix C): the first confirm attempt crashed before any statistic was written and was restarted with only a bug fix; and 19 of the 21 non-confirmed single-feature claims carry a stored reading of “untestable” that is wrong (they were tested and not detected), recomputed at render time, no verdict affected.

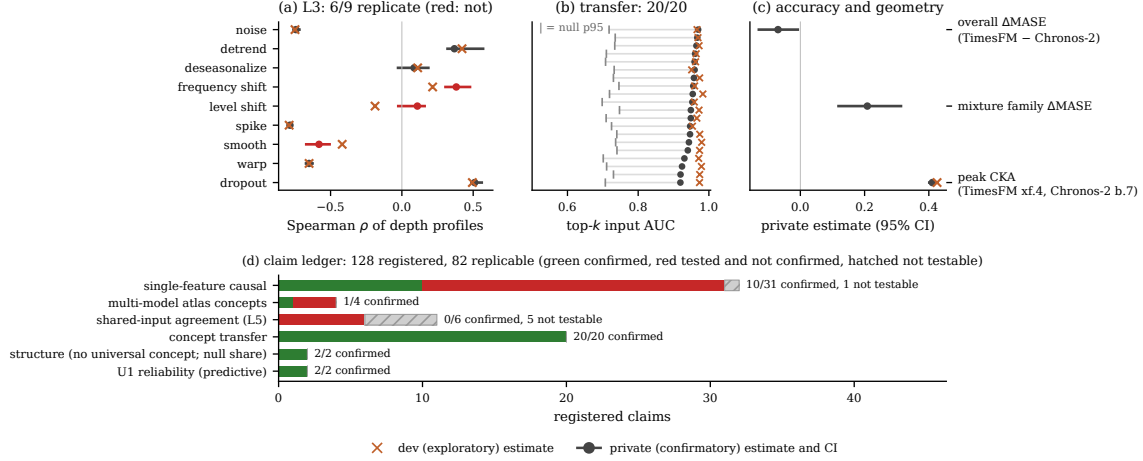


Figure 9: Claims re-tested on the sealed private split (971 series; panel a uses 800 of them). (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2 (red: the dev value is not inside the private interval); (b) the 20 concept-transfer claims with private null 95th percentiles; (c) accuracy and peak CKA; crosses: dev, dots and bars: private with 95% series-bootstrap intervals. (d) The ledger of registered concept claims by type: confirmed, tested and not confirmed, not testable (the one confirmed atlas claim is single-model). Source: `confirm/confirmation.json`.

8 Controls Change Conclusions

How can a reader know a system is better than a careful script running the same methods? Our answer is a record of cases in which the obvious analysis gave a well-formed, plausible and wrong answer, raised no error, and a specific control caught it. Table 6 lists fifteen; every control runs by default.

Three failure classes. *Interventions that never landed* (rows 1–4) produce flat curves or tables of zeros that read as strong negatives; the remedy (P3) is nearly free: self-patch exactly 0.0, cross-patch nonzero, two unpatched predictions identical, front end consistent with peers. *Nulls that were too weak* (rows 5–10) produce inflated positives that look like success (30 of 30 concepts transferring); the remedy is a null matched on the right confound, on the same rows, bounded from both sides, with the lower tail for claims of difference. Row 10 is caught by a known answer the system ships (Section 5.5): a forecaster whose concepts, decoys and a non-planted control layer are planted analytically, scored against a gate fixed in advance (false-positive rate ≤ 0.10 , sensitivity ≥ 0.5) on seeds held out from any tuning. The same device applied to the cross-model agreement rung shows it is specific but insensitive (MN-30), which is why the case study does not read its absences as differences. *Metrics that did not measure what they were named for* (rows 11–15) are caught only by validating on a known answer (P1): an untrained twin, a random rotation, known spans, a replicate seed. A grounded language model narrating features produced a new class of unsupported claim after each of four guard passes, while 6 of 6 deterministic sentences were correct (MN-06), so TSFM-LENS composes every report sentence deterministically. The table is a set of

RQ	Result (example run unless an ID is given)	Evidence class	Status	ID
1	194 causal features in 55 targets; 442 cells = $2.47\times$ empirical chance; Timer $1.02\times$	causal within-model	exploratory	MP-07, PM-17
1	9 families cover 99% of features but do not beat a shuffle null ($p = 0.68$)	descriptive	exploratory, 2 corpora	CA-10
1	Knob-specific concept mediation: 0/98, then 0/151	correlational	confirmed negative	MN-17
2	Peak CKA 0.426 (private 0.411) vs null 0.038	geometric	held-out confirmed	SH-01
2	7 similarity metrics agree only weakly ($W = 0.25$)	mixed	exploratory	SH-05/06
2	Untrained pair more similar than related trained pair	geometric	confirmed negative	MN-01
2	Concept transfer on inputs: 20/20	correlational	held-out confirmed (4 draws)	SH-18
2	No concept spans the panel (at most 3 of 7 models)	descriptive	held-out confirmed	SH-16
2	Atlas segregated by model ($p_{\text{below}} = 0.025$)	descriptive	exploratory	SH-22
2	Multi-model atlas concepts: 0/3	causal, compared	held-out not confirmed	CA-11
2	Same effect on same inputs: 7/216 scorable; 0/6 held-out; rung finds planted 0.4/0.0	causal on shared inputs	exploratory; not confirmed	SH-22, MN-30
3	Single-feature causal effects: 10/31	causal within-model	held-out partly confirmed	CA-11
3	84% of prominent features causally null (927 held-out)	causal within-model	held-out confirmed	MN-14
3	Feature effect $3.5\times$ below SAE reconstruction cost (earlier run)	causal within-model	exploratory	MN-15
3	Attention mass uncorrelated with head causality	causal vs. descriptive	exploratory	CA-04
3	Internals predict failure for the two weakest models only	predictive	held-out confirmed	MN-28
4	Depth-profile agreement, 6/9 corruptions	causal within-model	held-out partly confirmed	CA-01
4	Captured compute 14%–99.8% qualifies depth claims	descriptive	confirmed	DE-05
L0	Chronos-2 more accurate than TimesFM on <code>mixture</code> (0.209)	behavioral	held-out confirmed	PM-05

Table 5: Headline results with evidence class and status. Each ID is an entry in `FINDINGS.md` with exact numbers, reproduction command and artifact path.

counterexamples rather than a benchmark, and it lists only failures that were caught; the verification standard (`CONTRIBUTING.md`) requires each new test to be shown failing against a planted regression.

#	Analysis	Uncontrolled result	Controlled result	Built-in control	ID
<i>Interventions that never landed or were not identical</i>					
1	Skip lens, Chronos-2	MASE byte-identical (1.564247) at all 12 layers: “forecast formed at layer 0”	The head reads placeholder tokens the patch never writes; real reach by depth 0.737, 0.643, 0.282, 0.095, 0.000	reach probe; lens withheld when the head does not read patched positions	DE-03
2	Patching with bf16 cache	Aggregates within 0.66% of CI width of the corrected values	Self-patch was 0.00214, not 0; 71% of role assignments at one layer changed	self-patch must be exactly 0.0; clean cache with autocast off	MN-24
3	Sampled model (Sundial) as causal destination	Scorable in 1 of 71 tests: “Sundial is causally inert”	Null reseeded differently from its baseline; after the fix, scorable in 12 of 71	identical seeds across every compared pair of calls	PM-09
4	Scale equivariance, Sundial	Residual 38.43 context s.d. read as a model property; 100% flat forecasts on electricity	Adapter bypassed the checkpoint’s normalization; residual 0.030, flat share 0.08	frontend residual flagged against peers	PM-15
<i>Nulls that were too weak, or asked the wrong question</i>					
5	Concept transfer	Uniform random-subset null: 30 of 30 concepts transfer	Stratum-matched null: 56.3% forward, 46.1% reciprocal	null matched on benchmark strata	SH-14
6	Cross-model causal comparison	18 of 19 activation-matched roles “fire together, act differently”	Same series, own nulls, matched floors: 12 of 89 scorable tests act differently	shared-input agreement with matched floors	CA-06
7	Disagreement test	109 of 288 tests “act differently” (failing a q_{95} bar)	Lower-tail test: 9 act differently; 32 show no specific agreement	q_{05} required for disagreement	MN-18
8	Random-weight twins as causal floor	One floor per model	TimesFM’s twin is an exact identity stack yet reaches the forecast (0.187); Chronos twins are numerically dead ($< 10^{-7}$); floor exists for 1 of 6 pairs	within-run matched-feature floors; relative reach gate	PM-08
9	SAE names from ground truth	11 of 11 headline features named after corpus provenance	After residualizing, 82 of 88 cards (93%) ranked float-rounding noise (ρ 0.203 $\rightarrow$ 0.615)	residual-scale floor; control bounded from both sides	MN-05
10	Ablation-battery null	Planted forecaster: 17–25% of (feature, channel) cells on a layer with no planted effect clear the null (5 of 5 seeds); four-model case study: 293 causal features, 13 convergent concepts	Null removes the feature’s own per-token amount: false positives $\leq 0.3\%$, planted effects found at 0.70–1.00; same case study re-scored: 110 features ($3.1\times$ empirical chance), 0 convergent	planted-concept forecaster with a fixed gate; profile-matched null	MN-29
<i>Metrics that did not measure what they were named for</i>					
11	CKA and stitching for lineage	Related trained pair shows high similarity (CKA 0.734)	Untrained same-architecture pair is higher (0.878; $p = 0.0005$)	random-init twin comparison	MN-01
12	Unit-level correspondence	0.24–0.54, every CI excludes 0	Rotating both bases <i>raises</i> it (0.148 $\rightarrow$ 0.331)	rotation control on real activations	SH-19
13	Captured-depth chart	Chronos-T5 fully observed (1.0), TimesFM 0.851	Share of forecast FLOPs: Chronos-T5 14.4%, TimesFM 42.5%	FLOP-based coverage from the full forecast	DE-05
14	Span refusal statistic	“Diffuse” spans: refuses 3 of 5 correct adapters	Statistic grew with token count; peak-to-pedestal contrast accepts all (IoU 1.0)	gates normalized by attainable ceiling	MP-01
15	Single-seed SAE as gold standard	Layer selector beats random on 2 of 3 architectures	Verdict flips with the extraction seed; 3-seed average agrees across seeds	replicate SAE seeds	MP-04

Table 6: Fifteen documented cases in which an uncontrolled analysis gave a well-formed wrong answer and a built-in control changed the conclusion. Every control listed runs in the default configuration. Full accounts, including the exact commands, are in `FINDINGS.md` under each ID.

9 Using and Extending TSFM-Lens

A typical session installs the two packages, builds a benchmark and runs one configuration:

```
pip install -e . && pip install -e tsfm_lens
cd tsfm_benchmark && PYTHONPATH=. python3 example_runs/run_full.py \
    --config configs/example.yaml --out ./benchmark_out --references monash
cd ../tsfm_lens && python run.py --config configs/examples/medium_2model.yaml
```

The CLI adds a preflight check (`--doctor`), partial reruns (`--stages`, `--force`), checks for new checkpoints or library upgrades (`--check-alignment`, `--check-adapter`) and an environment diff (`--verify-provenance`). A CPU-only mock configuration runs every stage in minutes in continuous integration, and the worked example (`tsfm_lens/docs/worked_example.md`) walks through a report. The seven-model analysis records 21.1 GPU-hours of stage time, mostly SAE training and the concept battery; stages through L3 take about 1.4 hours. Of eight TSFM checkpoints probed, two (Timer, Time-MoE) load through the zero-code path and five with standalone packages need a hand-written adapter (MN-25), for which the template, checklist and guide exist (Appendix D). New report sections declare their artifact requirements, new claims must use the typed `Finding` record, and new configuration fields declare which artifacts they affect.

10 Limitations

Models. Seven transformer models with contiguous tokens; disjoint-lag tokens (Lag-Llama) get reduced scope. Timer’s features do not exceed chance, and with seven models, contrasts are confounded with architecture, size and training data. The analysis is univariate: Chronos-2’s cross-series group attention is not measured (PM-13), and Chronos-T5’s decoder and Sundial’s flow head are not captured.

Ground truth and confirmation. Ground truth is exact only for the synthetic tier and encodes our generators’ assumptions; 12 of 33 concept parts are dominated by one real-derived generator, and pretraining overlap cannot be verified. Held-out confirmation covers accuracy, geometry, perturbation agreement, transfer, single-feature causal effects, atlas structure and failure prediction, on one private split opened once (a second look needs a fresh epoch). Only 10 of 31 testable single-feature claims and no cross-model causal claim confirm; the agreement rung is insensitive (MN-30), so its failures are weak evidence of difference; 5 of 11 agreement claims and 1 of 32 causal claims could not be tested; and the 20 confirmed transfers come from two concepts.

SAE dependence, usability and completeness. Concept results depend on dictionary size, sparsity, revival and seed; the system measures this dependence but cannot remove it, and feature death tracks layer geometry (alive atoms scale with effective dimension with log-log exponent 1.02, $\rho = 0.58$ over 45 cells; MP-05). Readability for newcomers is a design intention; no user study measured comprehension. Controls catch only failures for which a known-answer case exists.

11 Conclusion

TSFM interpretability is harder than it looks: tokens mean different things in different models, there is no annotated ground truth, and causal effects are small enough that artifacts dominate unless controlled. TSFM-LENS addresses this with a controlled benchmark, measured alignment, null- and reach-gated causal concepts and a register-then-confirm protocol, behind one command and a report designed to be understandable by newcomers. On seven TSFMs, causal features act mainly on forecast level and seasonality and form a continuum; models select the same series for a concept, replicably, but no claim that they use it the same way survives the held-out test; and individual features account for little of the forecast: 43 of 82 replicable registered claims confirm. The fifteen caught failures show why these conclusions needed their controls.

Availability. Code (MIT), configurations, the example run and the findings ledger are at <https://github.com/DanK10097110/TSFM-Interp>; the dev branch holds the full research log and study drivers.

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Appendix A. Findings Ledger Index

Each identifier is an entry in `FINDINGS.md` giving exact numbers, models, evidence class, status, a reproduction pointer (configuration, command, artifact path), evidence class and status, and an interestingness score (1–5); negative results are included on equal terms. The table abridges each claim; evidence class and status are in the ledger. Prefixes: SH shared vs. unique; CA causal agreement; DE depth; PM per-model character; MP method positives; MN method negatives; BM benchmark trust.

Table 7: Ledger entries cited in this paper (score = interestingness, 1–5). *Section (§) or table (T) of this paper where the result is reported.

ID	Claim (abridged)	In paper*	Score
SH-01	Peak CKA 0.38–0.47 above shuffled-series nulls (0.02–0.16) across five architectures, including Timer via the zero-code adapter	§7, T5	5
SH-05	CKA and clustering AMI rank model pairs differently on three panels	T5	5
SH-06	Pair similarity splits into a geometric camp and a behavioral/causal camp; 7 metrics never pooled	T5	5
SH-14	Concept transfer on shared inputs vs. stratum-matched null: 56.3% forward, 46.1% reciprocal; uniform null gave false 30/30	T6	5
SH-16	No atlas concept spans the panel (at most 3 of 7 models; private atlas the same); “mostly convergent” did not survive the matched null	§7, T5	5
SH-18	Registered concept-transfer claims confirmed on private data: 19/20, 20/20 frozen, 20/20 on v2, 20/20 on the 7-model epoch 3	§7, T5	5
SH-19	Unit-level correspondence fails rotation controls (0.148 → 0.331); participation ratio as low as 1.54	§7, T6	5
SH-20	Per-target concepts mostly non-modular once a minimum cluster size is enforced: 50 of 56 targets at 7 models (11 of 13 earlier)	§5, §7	4
SH-22	Atlas segregated by model; all 7 “same causal effect” verdicts are Chronos-2/Bolt; rung insensitive	§7, T5	5

ID	Claim (abridged)	In paper*	Score
CA-01	Perturbation-fingerprint agreement is pair-specific; TimesFM/Chronos-2 replicates on private data	T5	4
CA-03	Seasonality circuit: 1 head (TimesFM) vs. 5 non-additive heads (Chronos-T5-Base)	\$7	4
CA-04	Attention mass at seasonal lags uncorrelated with causal importance ($\rho = 0.026, -0.118$)	\$7, T5	4
CA-06	On shared inputs with matched floors, most apparent cross-model disagreement disappears (18/19 $\rightarrow$ 12/89)	\$7, T5, T6	5
CA-07	Median level share of a causal feature's effect 0.535; most features shape-causal beyond a level-removed null	\$7	4
CA-10	Causal effect space is a continuum; families tile it but do not beat a shuffle null	\$7, T5	4
CA-11	Held-out: 10/31 single-feature causal claims confirm; no cross-model one does (atlas 0/3, agreement 0/6); 43/82 in all	\$7, T5	5
DE-01	Whole-layer patching hides a depth-intensifying recency effect in the last pre-horizon window	\$7	4
DE-03	Chronos-2 skip lens was a silent no-op (identical MASE at 12 layers); corrected reach 0.737 $\rightarrow$ 0.000	\$5, \$7, T6	5
DE-04	Chronos-T5 encoder ends at depth 0.478 on a whole-stack axis; 50% comparable overlap with TimesFM	\$5, \$7	5
DE-05	Captured share of forecast FLOPs: 13.9% (Chronos-T5-Base) to 99.8% (Timer) at 7 models (14.35% to 97.5% on four)	\$4, \$7, T5, T6	5
DE-08	Layer selection depends almost entirely on the selector (0/3 robust layers)	\$7	4
PM-05	Chronos-2 most accurate and best observed; winners vary by family; held-out on v2 and on 7-model epoch 3	T5	4
PM-07	Models differ $\sim 5.4\times$ in reversion to the mean; flat forecasts driven by weakly predictable data	\$2	4
PM-08	Random-weight twins are not one null: identity stack (TimesFM) vs. dead forecasts (Chronos-2/Bolt)	T6	5
PM-09	Sundial scorable as causal destination 1/71 $\rightarrow$ 12/71 after seeding fix	T6	3
PM-11	Causally load-bearing TimesFM random-walk concept fails seed stability	\$7	3
PM-13	Chronos-2 group attention unmeasured by design	\$9	3
PM-15	Sundial adapter bypassed checkpoint normalization; residual 38.43 $\rightarrow$ 0.030	\$7, T6	4
PM-17	Timer's SAE features do not clear the ablation null above chance (1.017 $\times$; 0 of 6 targets)	\$7	3
MP-01	Token spans can be measured (IoU 1.0 on 6 checkpoints); zero-code adapter follows	\$4, T6	5
MP-02	Alignment gates must be normalized by their attainable ceiling	\$4	3
MP-03	AuxK revival recovers dead SAE atoms 10–13 $\times$ on real activations	\$5	4
MP-04	Single-seed SAE gold ranking flips a bake-off verdict; 3 seeds fix it	\$5, T6	5
MP-05	SAE feature death set by layer geometry (exponent 1.02), not recipe	\$9	5
MP-06	Ground-truth alignment of SAE features is real only over a permutation null	\$7	3
MP-07	Ablation battery discriminates: 2.47 $\times$ empirical chance at 7 models (Timer 1.02 $\times$, Bolt 5.70 $\times$)	\$7	4
MP-09	Dev and private splits exchangeable (energy $p = 0.51$; TOST 21/22)	\$4	3
MN-01	Similarity cannot detect lineage: untrained pair 0.878 > trained related pair 0.734	\$7, T5, T6	5
MN-05	Residualizing on provenance made 93% of headline cards float-rounding noise	T6	5
MN-06	A grounded small LLM narrator fabricates comparisons; guards do not converge	\$8	5
MN-07	Causal fingerprints are nearly one-dimensional (PR 2.09)	\$7	4
MN-14	0.835 of 927 prominent SAE features are causally null on private data (dev 0.810; 64.4% under the legacy null)	\$7, T5	4
MN-15	SAE reconstruction moves the forecast $\sim 3.5\times$ more than removing a feature	\$7, T5	4
MN-16	"Reconstruction beats clean forecast" replicated in 1/5 seeds; retracted	\$8	4
MN-17	Generator-counterfactual concept mediation: 0/98 and 0/151	\$7, T5	5
MN-18	Absence of agreement is not disagreement: 109/288 $\rightarrow$ 9	\$5, T6	4
MN-19	"Same inputs" must be top- k overlap, not whole-series correlation	\$5	4
MN-21	Reviving dead features hurts ground-truth alignment in TimesFM at every depth	\$7	5
MN-23	MASE denominator floors out on intermittent series	\$5	3
MN-24	A precision mismatch left aggregates intact but reorganized 71% of a partition	T6	4
MN-25	Zero-code adapter resolves 2 of 8 probed TSFM checkpoints	\$9	3
MN-28	Internals beat a free failure-prediction baseline only for the two weakest models; routing does not help	\$7.6	4
MN-29	Legacy ablation null fails a planted-concept gate (FPR 0.168–0.248); profile-matched null passes (≤ 0.003)	\$5.5, T6	5
MN-30	Agreement rung is specific (0/9 false "same") but insensitive (0.4 and 0.0 of seeds)	\$5.5, \$7	5
BM-05	Counterfactual identity check caught a corruption-replay bug (244/298 $\rightarrow$ 0/1113)	\$4	2

Appendix B. Statistical Details

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}}$ resampled means $\bar{d}_b^*$, the interval is the percentile interval and $p = \min(1, \max(\frac{1}{B}, 2 \min\{\frac{1}{B} \#(\bar{d}_b^* \leq 0), \frac{1}{B} \#(\bar{d}_b^* \geq 0)\}))$ (analysis/stats.py; floor per Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Holm: $\tilde{p}_{(i)} = \max_{j \leq i} \min(1, (m - j + 1)p_{(j)})$ (Holm, 1979); with $p \geq 1/B$ the smallest attainable value is m/B , which the ledger reports per family, flagging $m/B > \alpha$. BH: $\tilde{q}_{(i)} = \min_{j \geq i} \min(1, \frac{m}{j} p_{(j)})$ (Benjamini and Hochberg, 1995).

Profile-matched ablation null. For feature i with rows R_i and $J = 16$ unit directions u_j , the null removes at every token t the amount $|z_i(t)| \|w_i\| u_j$ (the legacy null removed one batch-averaged amount $\bar{a} W_d u_j$); channel c clears if $\frac{1}{|R_i|} \sum_{r \in R_i} |\Delta_{i,r}^{(c)}| > q_{0.95} \{|\Delta_{u_j,r}^{(c)}| : j \leq J, r \in R_i\}$. A null with no spread is reported as degenerate and not scored.

Transfer. The statistic is $\max_f \text{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k -sets, so the feature search is inside the null. The reverse leg scores the chosen feature’s own top- k by the source activation. $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (1 + n)$; a registered claim uses $\max(p, p_{\text{rev}})$ (intersection-union), Holm across claims.

Shared-input agreement. On U , statistic (i) is the Spearman concordance of the two sides’ signed level effects across series, and (ii) the cosine of their level-removed shape vectors, each channel in its own null units. Each side’s floor ablates random feature sets matched to its activation on U and re-scores against the other side’s measured effect; agreement exceeds both floors’ $q_{0.95}$, “acts differently” falls below both $q_{0.05}$.

Minimum detectable effect. Before the private result is read, recentred per-series differences are shifted by a candidate effect and re-tested with the same bootstrap; bisection finds the smallest shift detected with power 0.8 (`analysis/power.py`).

Appendix C. Reproducibility

`examples/panel7_v2` holds the configuration, the report and every stage’s JSON/parquet artifacts plus the confirmation (activations and SAE weights omitted for size); `paper/figures/make_figures.py --run DIR` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/panel7_v2.yaml` reproduces the run (add `--stages report` to re-render, and `--verify-provenance runs/panel7_v2_dev` to diff environments). The private split (v2 epoch 3, 971 series) was opened once (2026-10-02), so re-confirming requires a fresh epoch. Two events are disclosed. A first epoch minted for this look (v2 epoch 1) was rejected before any model saw it, because 716 of its 961 series equalled series of a split an earlier confirmation had consumed (epoch seeds depend only on the seed and epoch number). And the first confirm attempt crashed after every private battery finished, while assembling the claim ledger (a fixed family map lacked the failure-prediction family), before any statistic was written: `confirmation.json` did not exist, no private number was read, the registry hash and every frozen specification were unchanged, and the restart differed only by the crash fix. It is one test of the registry run twice for compute reasons; the artifact records `repeated_look: false`. Environment, as recorded in the run’s `run_manifest.json`: Python 3.12.13, numpy 2.1.0, PyTorch 2.12.0 (CUDA 12.9 build), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7 (zarr 3 stores read back empty under the v2 API), pandas 2.3.3, scikit-learn 1.7.2; load-bearing pins and their reasons are in `DEPENDENCIES.md`. Tests (collected; not re-run for this revision): 2,520 tests (`tsfm_lens`) and 125 (`tsfm_benchmark`);. Recorded stage wall-clock time on one RTX A5000: 75,900 s in total, of which `sa` 14,614 s and `concepts` 55,608 s (shared-input agreement 45,189 s); every other stage under 33 minutes.

Appendix D. Adapter Pitfalls: a Worked Case

Sundial z-scores inputs inside `generate()`, but `forward()` defaults to no normalization. `generate()` was broken on the installed transformers, so the adapter called `forward()` and fed raw-scale input (PM-15). The `frontend` stage measured a scale-equivariance residual of 38.43 context standard deviations (peers ≤ 0.009) and 100% flat forecasts on electricity, and the residual was first recorded as a model property. The adapter now normalizes in `prepare()` with the checkpoint’s exact rule and denormalizes in `predict()`, so capture, prediction and patching see identical inputs: residual 0.030, flat share 0.08; the same defect in `generic_hf` on Timer fell from 49.73 to 0.020. The report now flags any residual above $\max(1, 10 \times \text{the peers’ median})$. Lesson: an adapter calling a model below its public API must reproduce everything that API does to input and output. The guide (`tsfm_lens/docs/ADDING_A_MODEL.md`) turns this and the other pitfalls (identical seeds for compared calls of sampled heads, clean-cache precision, no batch chunking, measured spans, reach controls) into a checklist.

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